DCA0305

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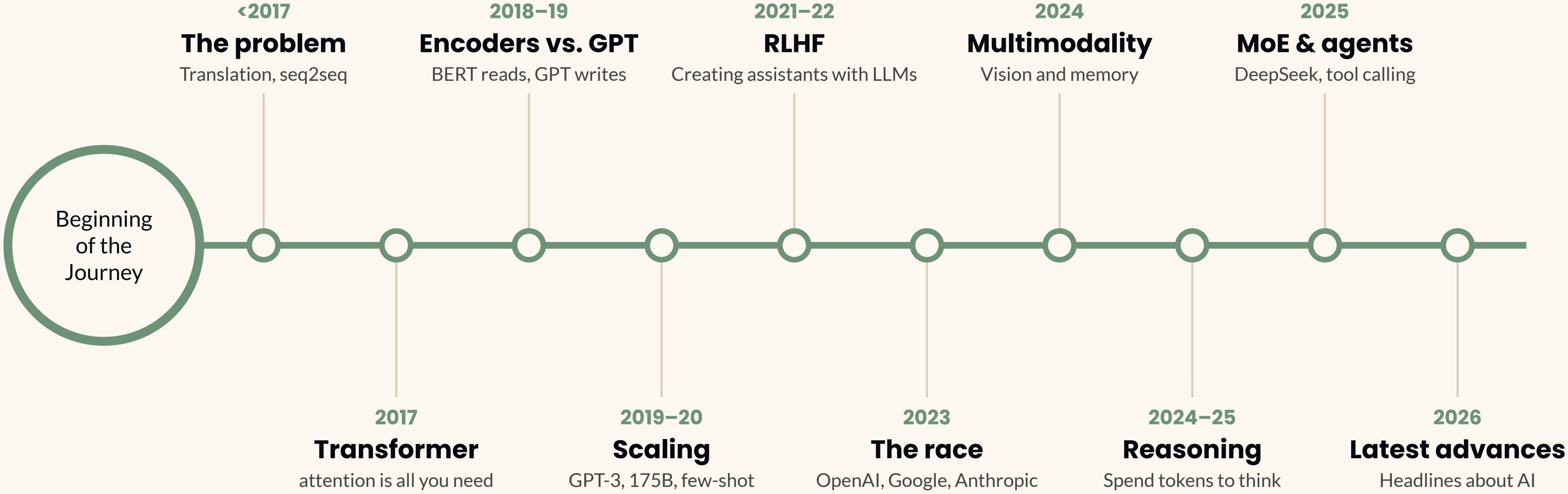
A Brief History of LLMs

From one word at a time to agents · 2014 → 2026

The Transformer architecture is nine years old.
ChatGPT is almost four years old.

June 2017 · November 2022


Beginning of the Journey
<2017
The problem
Translation, seq2seq
2017
Transformer
attention is all you need
2018–19
Encoders vs. GPT
BERT reads, GPT writes
2019–20
Scaling
GPT-3, 175B, few-shot
2021–22
RLHF
Creating assistants with LLMs
2023
The race
OpenAI, Google, Anthropic
2024
Multimodality
Vision and memory
2024–25
Reasoning
Spend tokens to think
2025
MoE & agents
DeepSeek, tool calling
2026
Latest advances
Headlines about AI

Before 2017: translation, by counting

Statistical MT

le chat s'est assis

Phrase tables
counts from large corpora

le chat	→	the cat	72%
		cat	18%
		this cat	6%
s'est assis	→	sat	61%
		is seated	24%
		sat down	10%

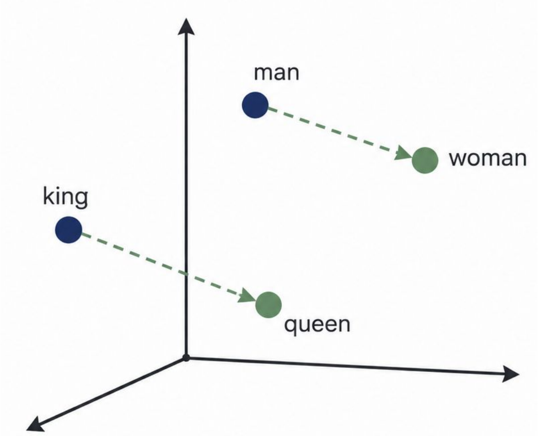
n-gram model
P(sat | the cat) ↑

the cat sat

Google Translate
2006 → 2016

Word2vec · 2013

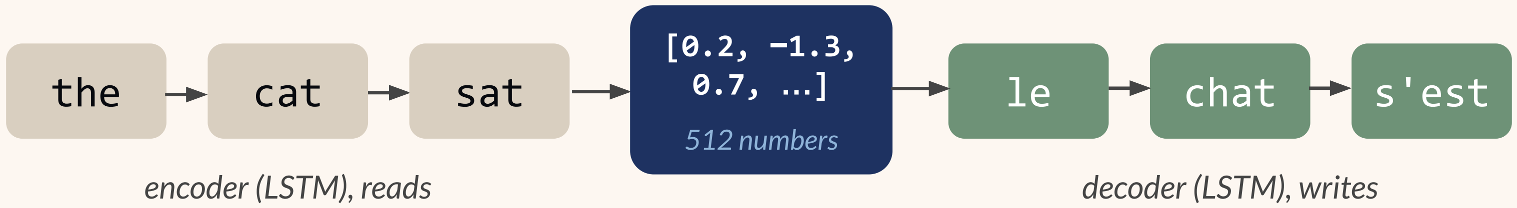
Mikolov et al. · Google



Similar meaning → nearby vectors

Meaning becomes arithmetic

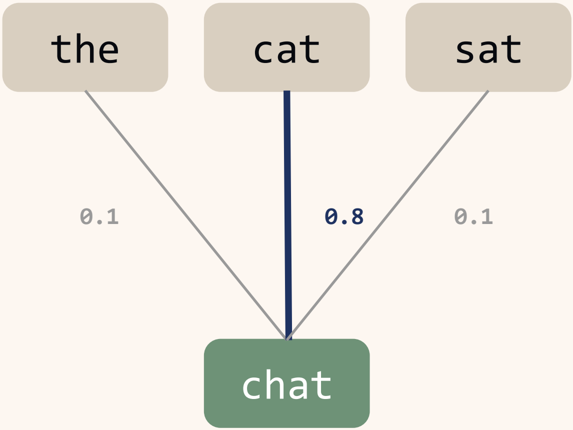
2014: squeeze the sentence into one vector



One fixed-size vector.
5 words or 50 words, same 512 numbers.

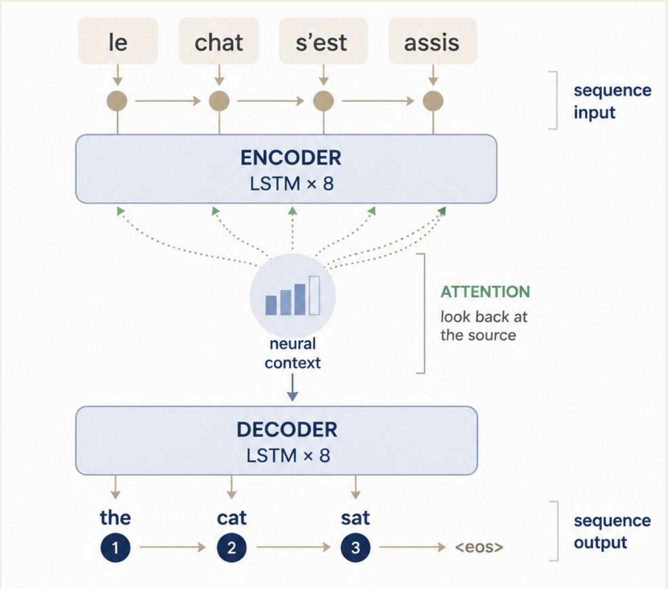
seq2seq · Sutskever, Vinyals & Le (Google) · NeurIPS 2014

2015: attention, as a patch



The decoder weights every input word, every step.

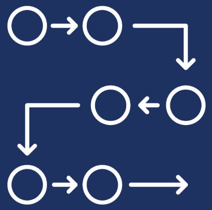
GNMT · 2016 *Google Translate goes neural*



Bahdanau, Cho & Bengio · ICLR 2015

2017: three problems on the table

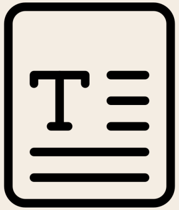
1



Sequential

Word n waits for word n−1
GPUs mostly idle
Vision went parallel in 2012

2



Long paths

Word 1 to word *n* crosses *n* steps
The signal fades
Long dependencies are lost

3

Cost

Low training caps size and data
Weeks per experiment
No way to scale

2017: Attention Is All You Need

arXiv:1706.03762v1 [cs.CL] 12 Jun 2017

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

1 Introduction

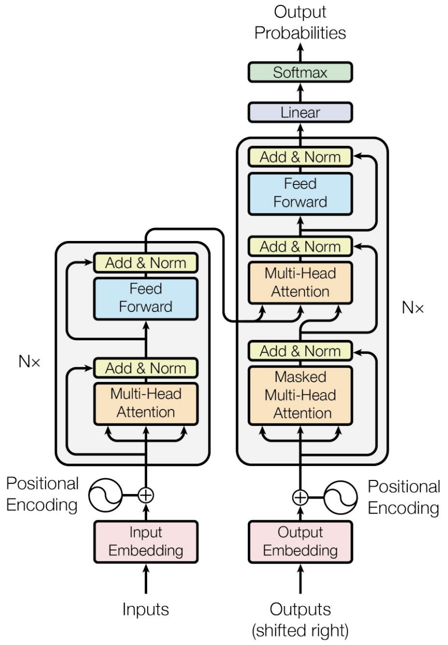


Figure 1: The Transformer - model architecture.

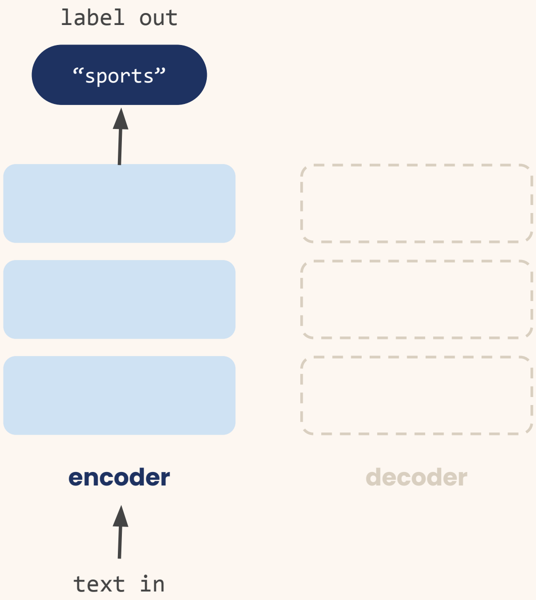
Encoder
Reads the input, all words at once

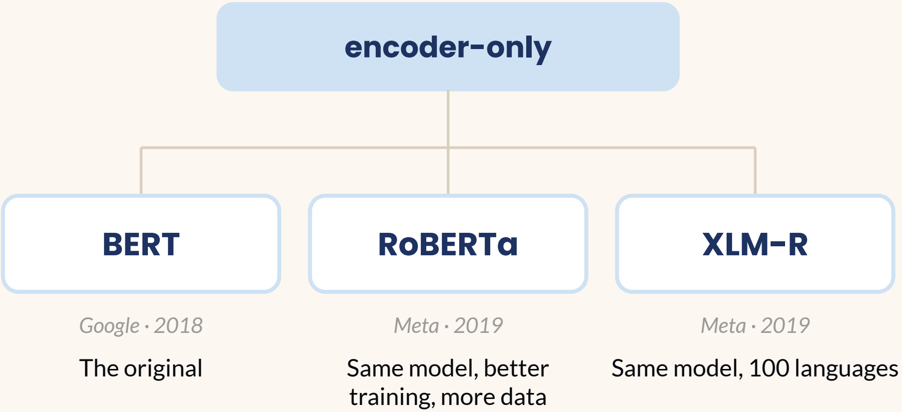
Decoder
Writes the output, one token at a time

3.5 days · 8 GPUs
BLEU 28.4 on WMT'14 En→De
built for translation

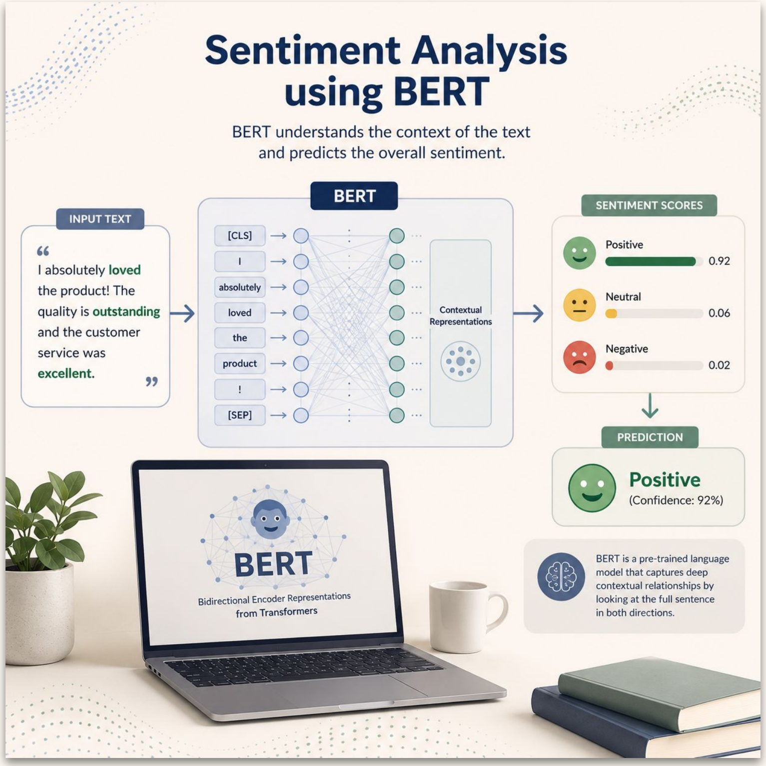
Vaswani et al. · Google · June 2017

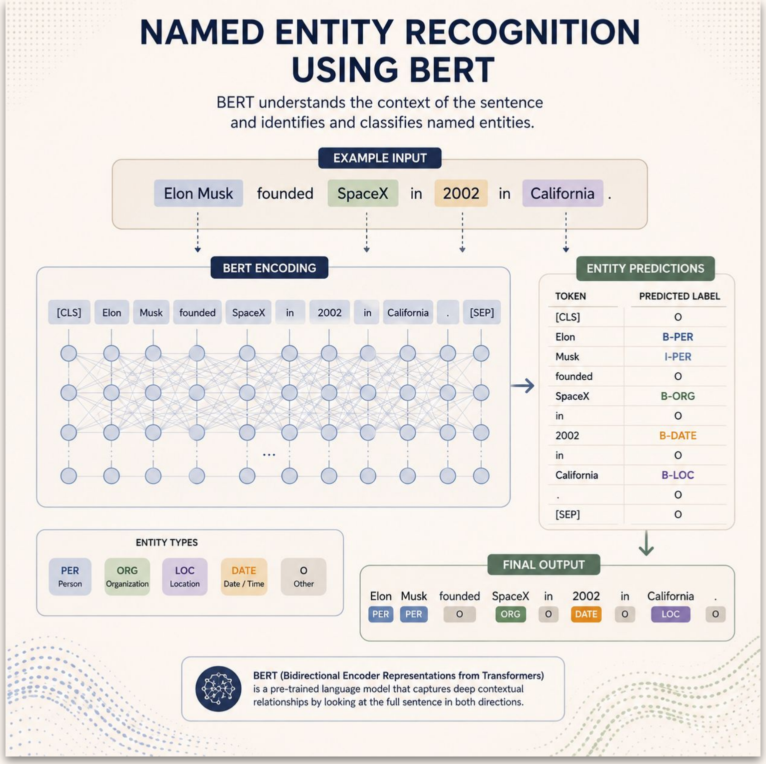
Encoders vs. GPT: models that read



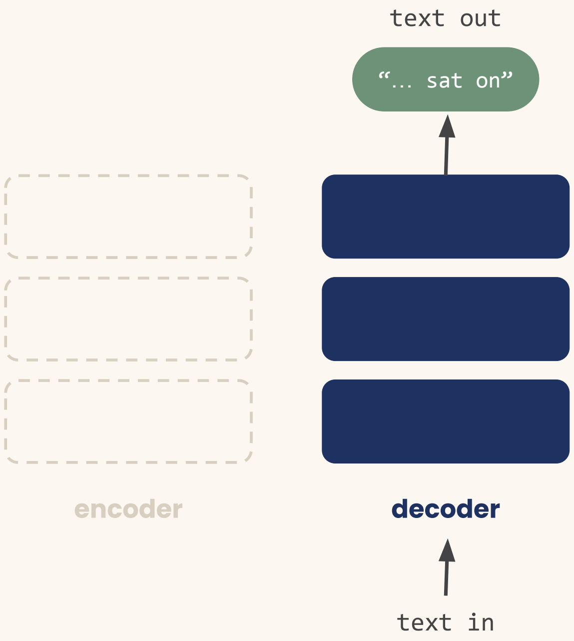


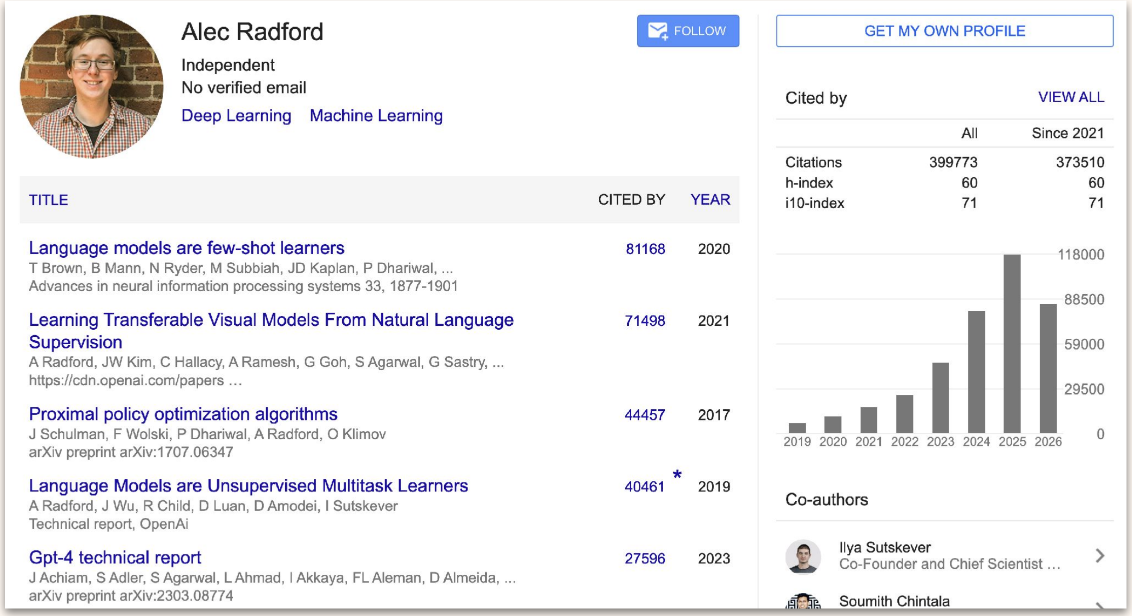
Encoders vs. GPT: models that read





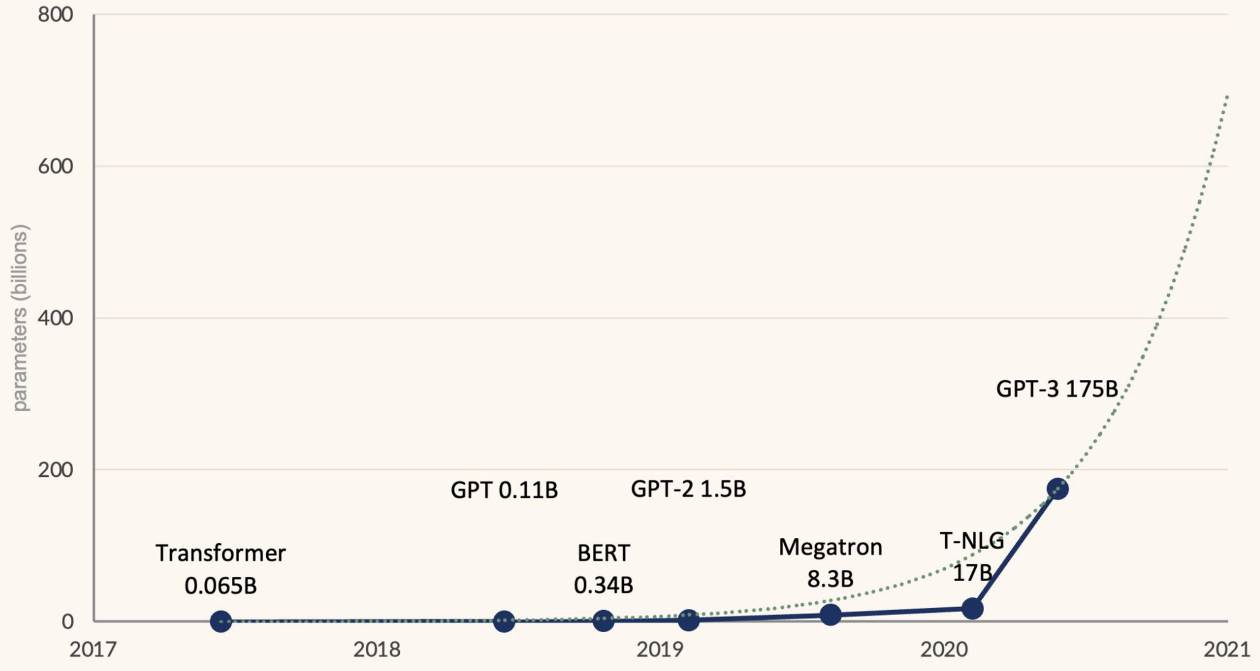
Encoders vs. GPT: the contrarian bet





Alec Radford · led GPT (2018) and GPT-2 (2019)

Scaling: NLP’s Moore’s law



~10× per year

● dense models
··· ×10 per year

GPT-3: 100× GPT-2 in one year
570 GB of text · ~500B tokens

Scaling: the task lives in the prompt

prompt

```
A “whatpu” is a small, furry animal native to Tanzania. An example of a
sentence that uses the word whatpu is:
We were traveling in Africa and we saw these very cute whatpus.

To do a “farduddle” means to jump up and down really fast. An example of a
sentence that uses the word farduddle is:
```

completion

```
One day when I was playing tag with my little sister, she got really
excited and she started doing these crazy farduddles.
```

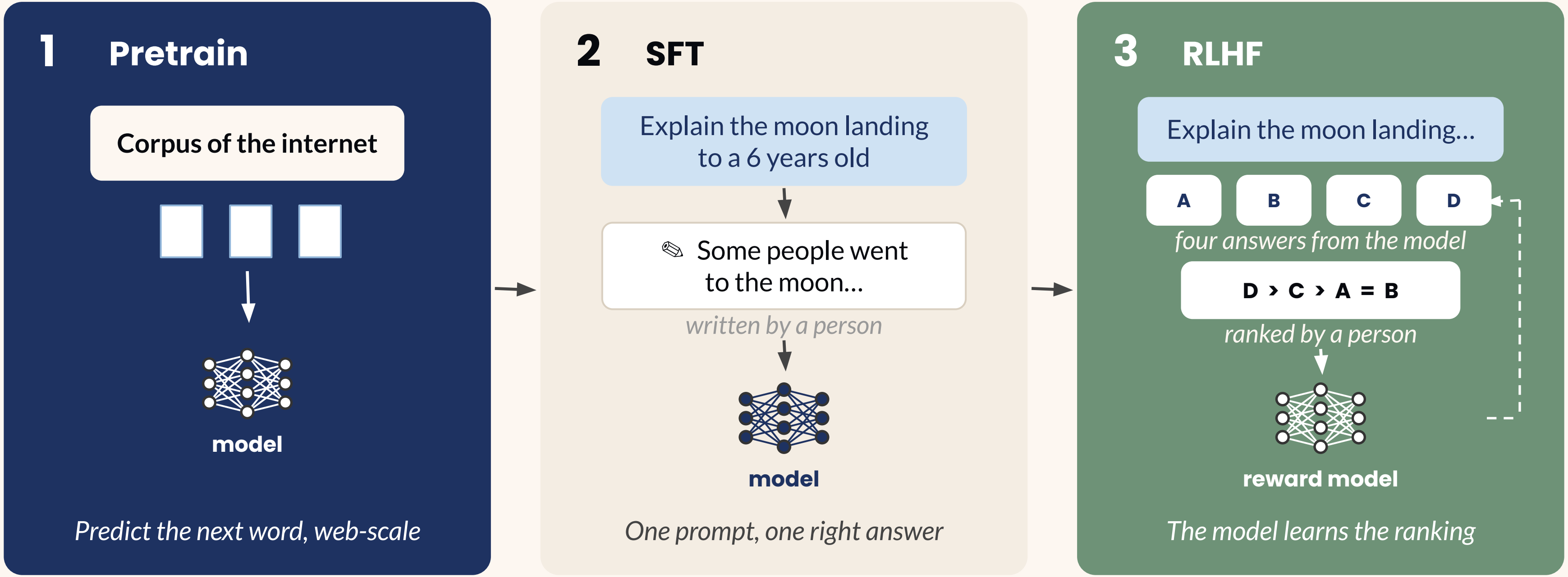
No retraining

One example of the shape, definition then sentence.

It produces another with a word that **does not exist.**

GPT-3 · “Language Models are Few-Shot Learners”, 2020.

RLHF: teaching it to talk

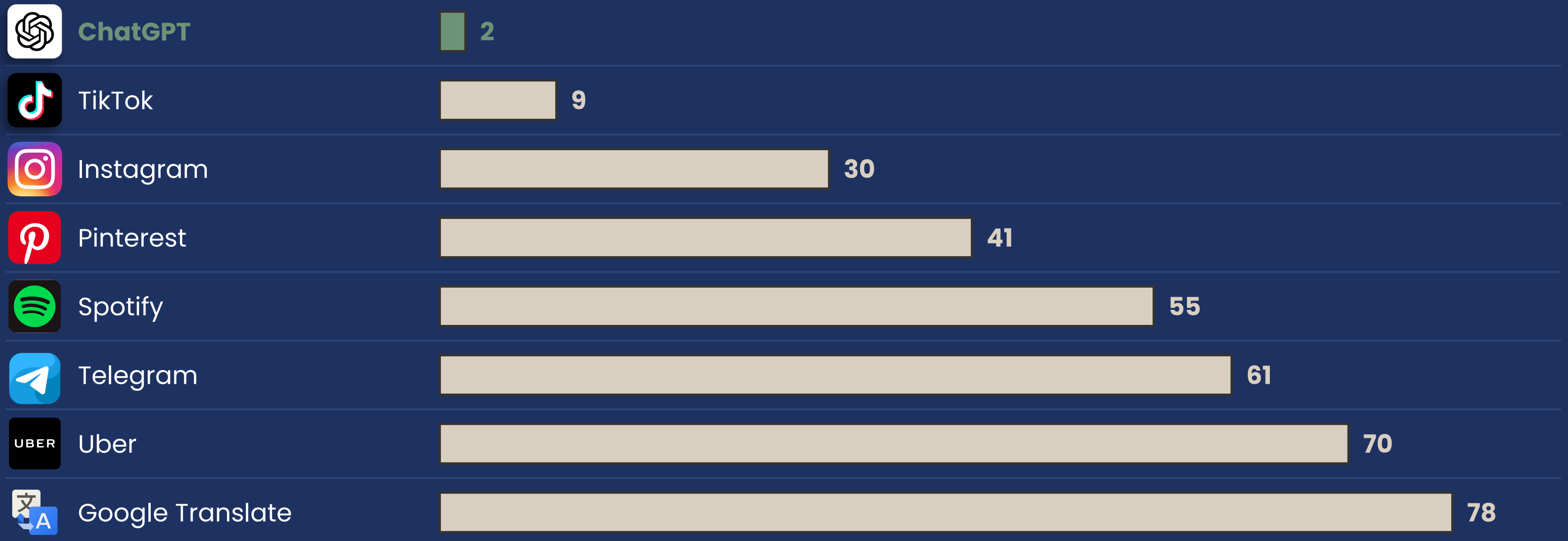


InstructGPT (Ouyang et al., 2022)

Nov 2022

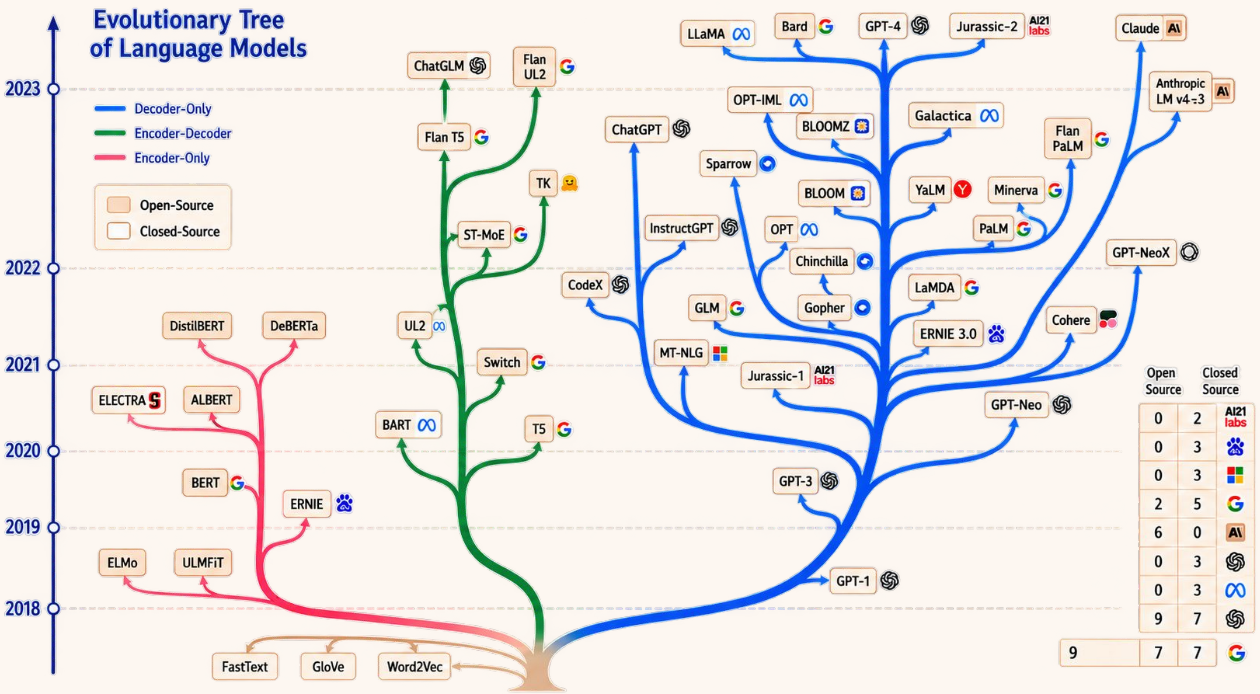
ChatGPT = InstructGPT + engineering

months to reach 100M monthly users



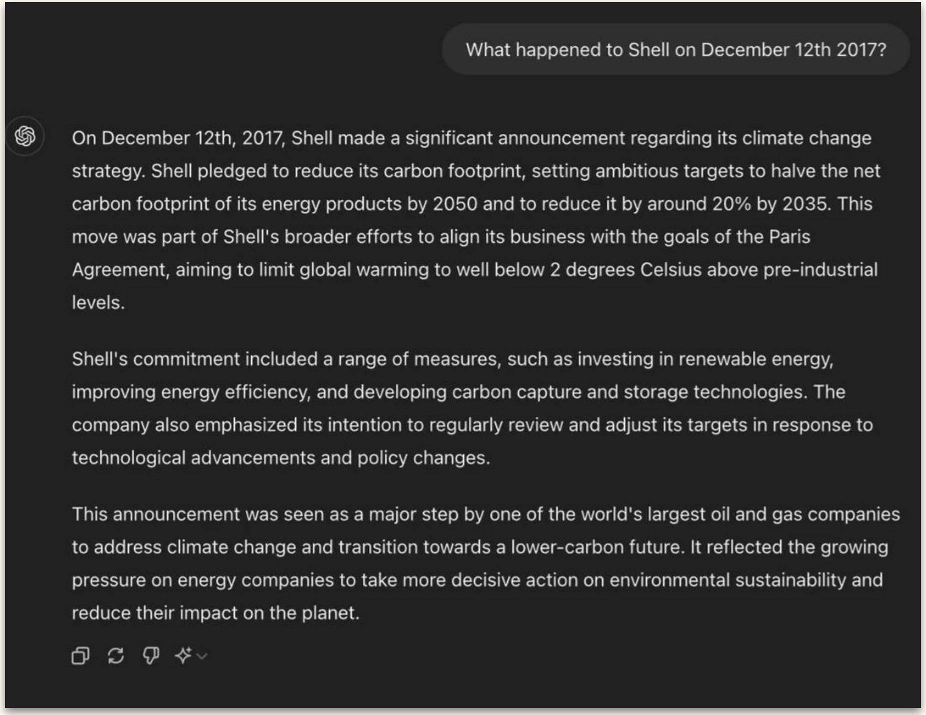
source: UBS, Feb 2023 (reported by Reuters / Yahoo Finance)

The race: everyone joins



Adapted from “Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond”, 2023.

The race: limitations of ChatGPT



What happened to Shell on December 12th 2017?

On December 12th, 2017, Shell made a significant announcement regarding its climate change strategy. Shell pledged to reduce its carbon footprint, setting ambitious targets to halve the net carbon footprint of its energy products by 2050 and to reduce it by around 20% by 2035. This move was part of Shell's broader efforts to align its business with the goals of the Paris Agreement, aiming to limit global warming to well below 2 degrees Celsius above pre-industrial levels.

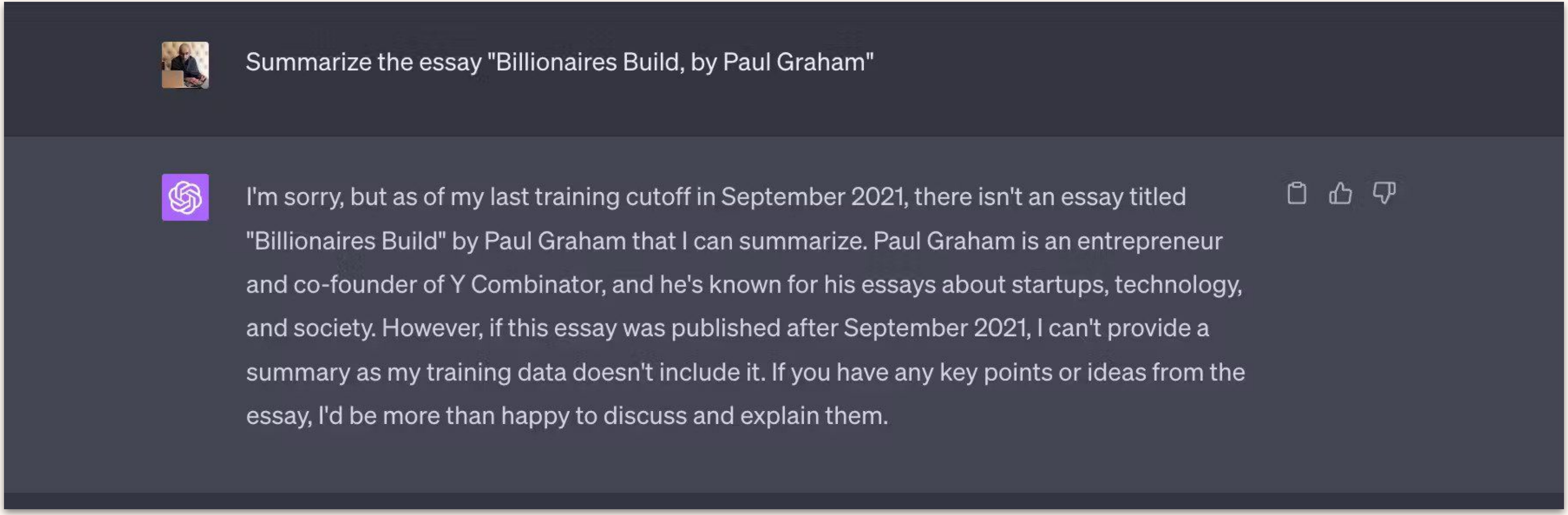
Shell's commitment included a range of measures, such as investing in renewable energy, improving energy efficiency, and developing carbon capture and storage technologies. The company also emphasized its intention to regularly review and adjust its targets in response to technological advancements and policy changes.

This announcement was seen as a major step by one of the world's largest oil and gas companies to address climate change and transition towards a lower-carbon future. It reflected the growing pressure on energy companies to take more decisive action on environmental sustainability and reduce their impact on the planet.

Factual hallucination

Correct date of the announcement. Source: United Nations Climate Change

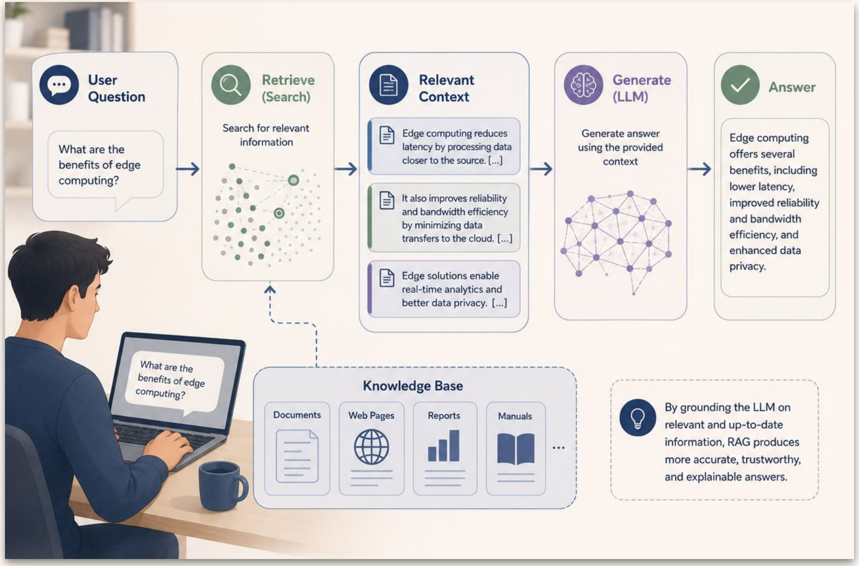
The race: limitations of ChatGPT

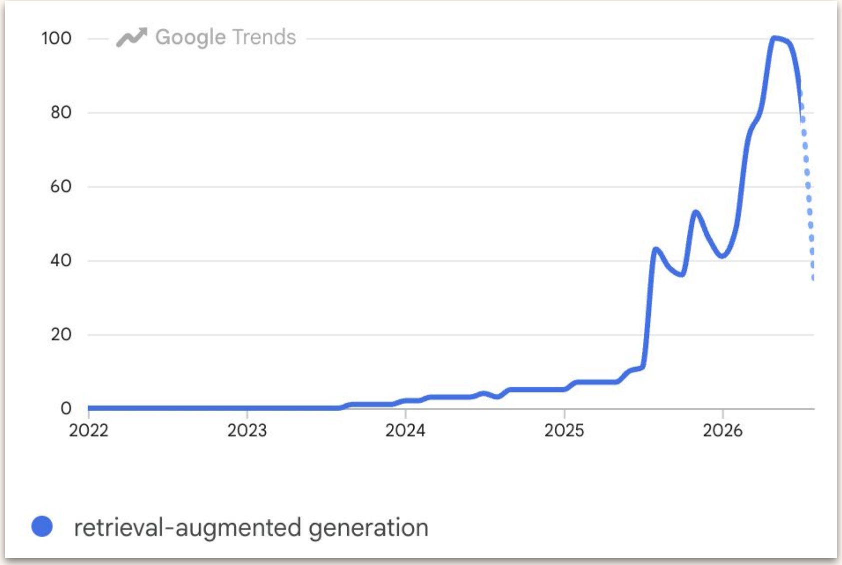


Knowledge cut-off

The race: your data in the prompt

RAG · Retrieval-Augmented Generation





Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks", NeurIPS 2020

Multimodality: vision and memory

1,000,000

Tokens in one prompt · Gemini 1.5 Pro · Feb 2024

1 h	11 h	30k	700k
of video	of audio	lines of code	words

402 pages of Apollo 11 transcripts, one prompt.

4k
2022 · ChatGPT

32k
2023 · GPT-4

128k
2023 · GPT-4 Turbo

200k
2023 · Claude 2.1

1M
2024 · Gemini 1.5

Example of GPT-4 visual input:

User What is funny about this image? Describe it panel by panel.

Source: https://www.reddit.com/r/hmmm/comments/ubab5v/hmmm/

sees images ✓
generates images ✗

sources: blog.google, Gemini 1.5 launch (Feb 2024) · GPT-4 technical report (OpenAI, Mar 2023)

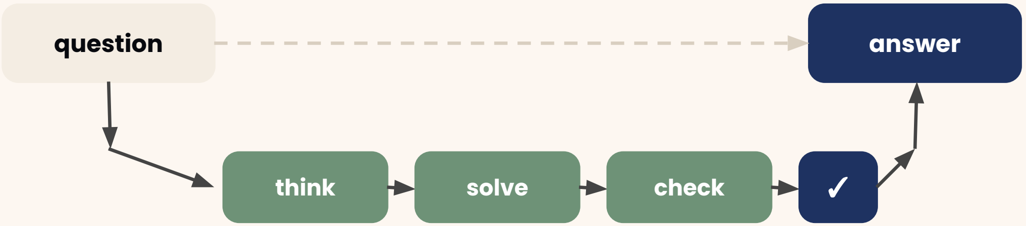
Reasoning: spend tokens to think

13% → 83%

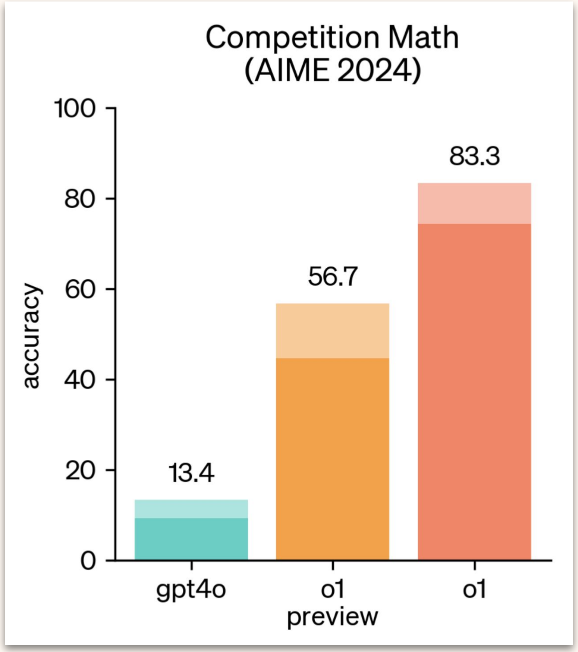
GPT-4o vs o1 on the AIME (the IMO qualifying exam) · Sep 2024

Same lab, same year. The difference is thinking before answering.

2020: straight to the answer



2024: think first, then answer · more tokens, more time, better on math and code



Source: OpenAI's blog · Learning to reason with LLMs, 2024.

MoE & agents: the DeepSeek bomb

−$589B

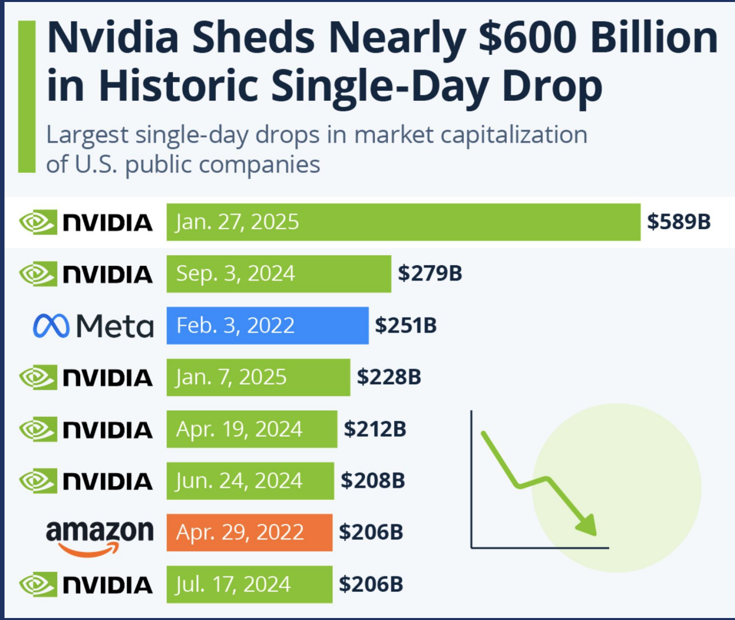
Nvidia became the company with the biggest single-day loss ever.

One week earlier, NVIDIA had just overtaken Apple as the most valuable company in the world.

Mon 20 Jan
DeepSeek publishes R1
open weights, MIT licence

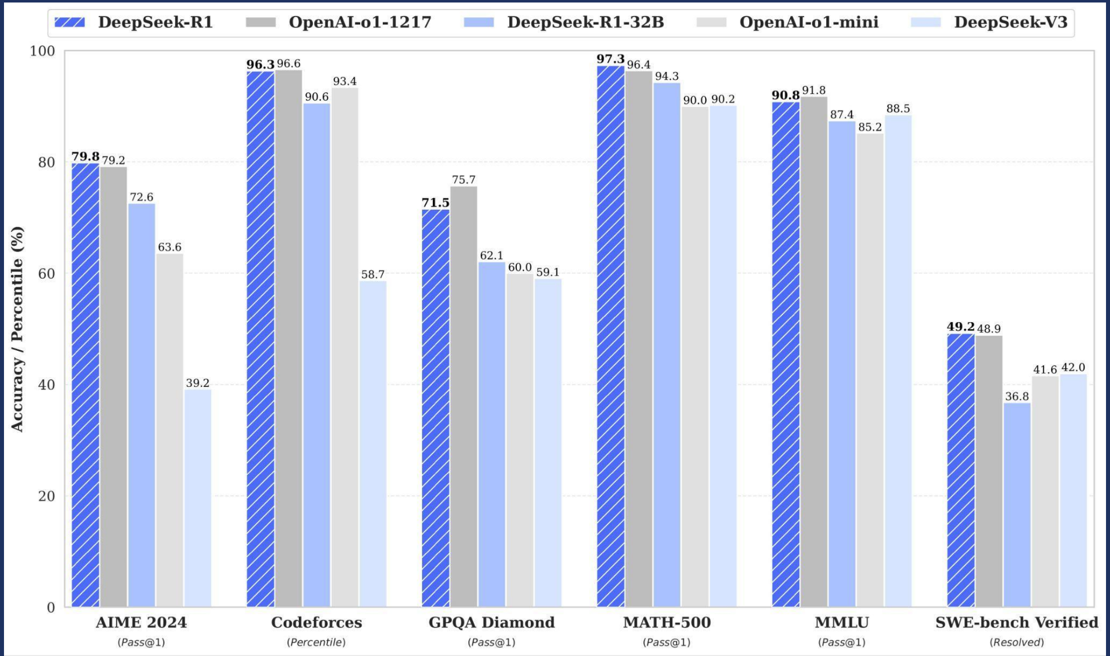
The weekend
App and paper go viral
The DeepSeek app reaches #1 on the US App Store.

Mon 27 Jan
Markets open
NVIDIA −17% · Nasdaq −3.1%



Sources: CNBC, Forbes, Statista (27–28 Jan 2025)

MoE & agents: why it happened



R1 vs o1 · AIME 79.8 / 79.2 · GPQA 71.5 / 75.7 · Codeforces 96.3 / 96.6 · SWE-bench 49.2 / 48.9

o1-level

A tie with the best closed model in the world, on math, code and science.

~10% of the cost

$5.6M declared for V3 vs >$100M estimated for GPT-4.

open weights, MIT

Anyone can download it, run it, fine-tune it, sell it

O1
Dense model. All parameters are active.

DeepSeek
Mixture of Experts (MoE). 2 out of 8 are active.

MoE & agents: one model for pixels

Sam Altman
@sama · Follow

the chatgpt launch 26 months ago was one of the craziest viral moments i'd ever seen, and we added one million users in five days.

we added one million users in the last hour.

3:11 PM · Mar 31, 2025

56.8K Reply Copy link

Read 2K replies

2024

LLM — *sees images*

diffusion — *makes images*

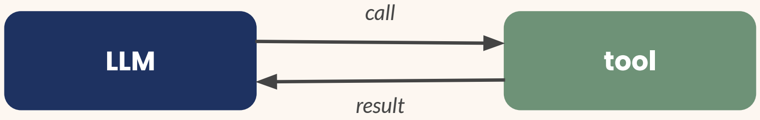
→

2025

one model — *sees and makes*

Ghibli style GPT 4o image generation

MoE & agents: tool calling



Call a tool, read the result, decide the next step.

Run locally

Ollama

Any open model on your laptop

Code

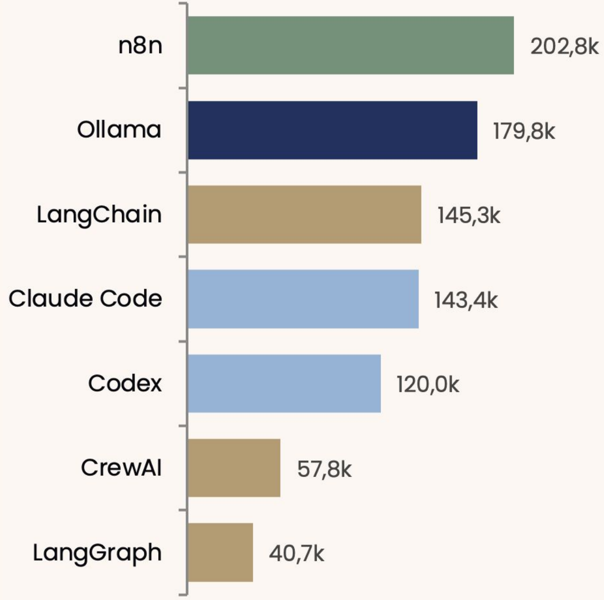
LangChain · LangGraph · CrewAI

Chains, graphs, crews of agents

No code

n8n · Langflow

Drag boxes, wire them



GitHub stars, 30 Aug 2026

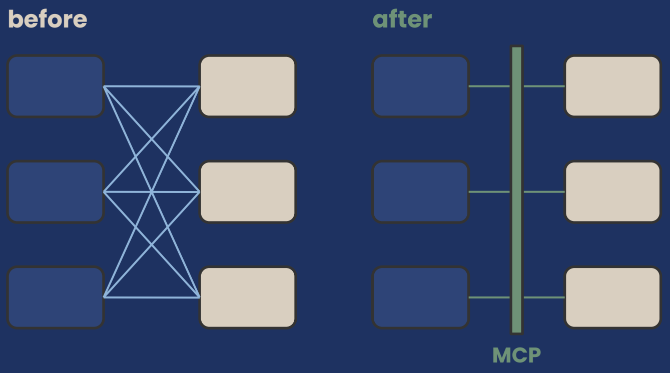
MoE & agents: a USB port for tools

6 months

Nov 2024
Anthropic publishes MCP
Open spec

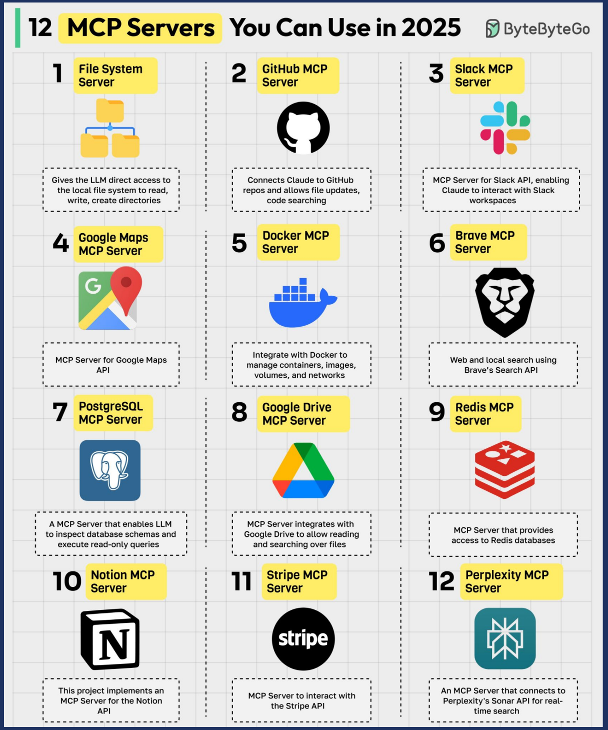
Mar 2025
OpenAI adopts it
ChatGPT, Agents SDK

Apr 2025
Google adopts it
and proposes A2A, agent ↔ agent



N × M integrations

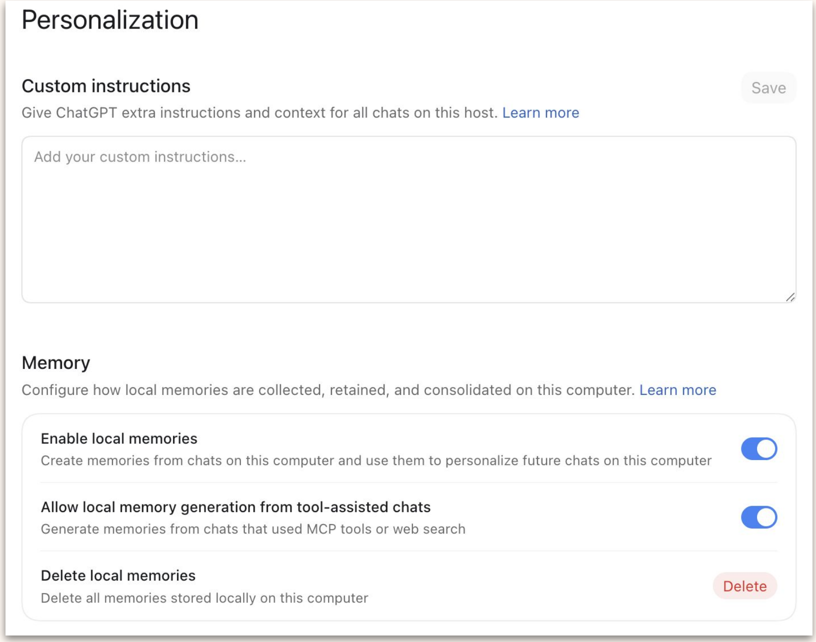
N + M



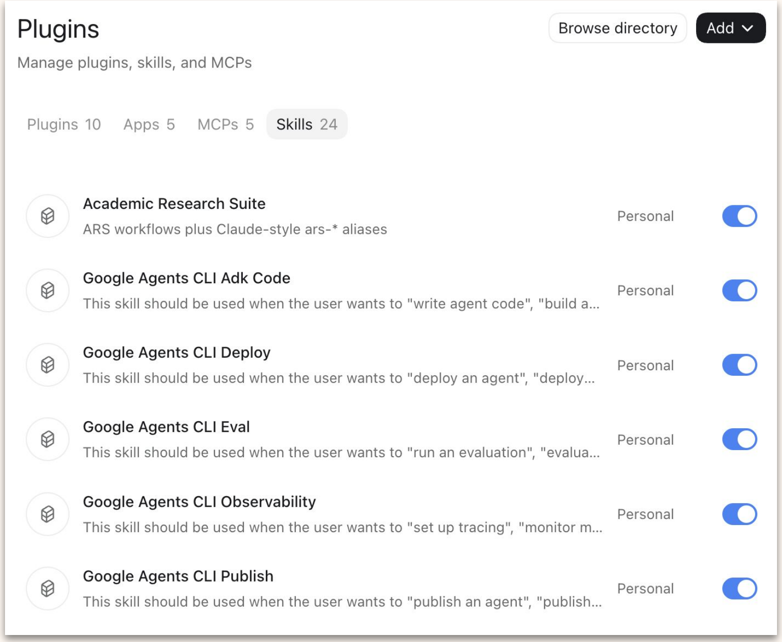
Source: ByteByteGo, 2025.

MoE & agents: memory and skills

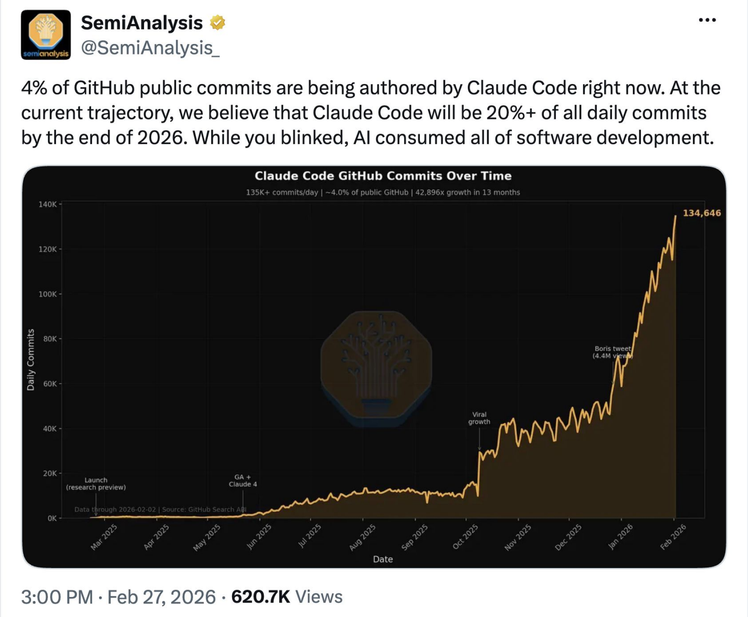
Memory



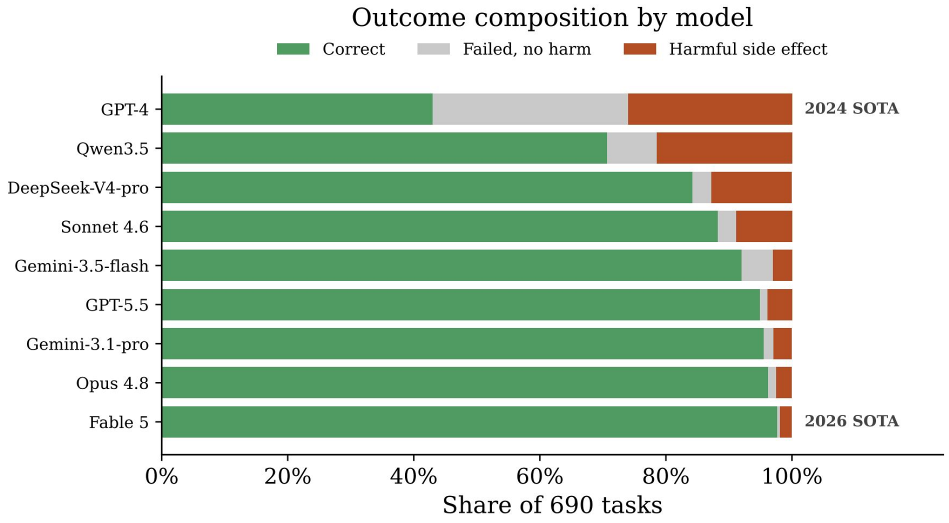
Skills



Latest advances: the headlines (1/2)



x.com/SemiAnalysis_/status/2027443723362042308



WorkBench Revisited, 2026. Source: arxiv.org/pdf/2606.13715

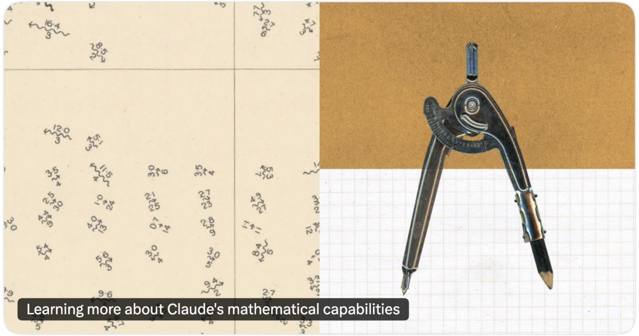
Latest advances: the headlines (2/2)

Anthropic @AnthropicAI

We asked an unreleased research version of Claude to take a stab at the Riemann hypothesis.

It didn't solve it, but it did make strides on a related problem: it increased the lower bound for the fraction of zeros of the Riemann zeta function that satisfy the hypothesis from 41.6% to 67.2%.

anthropic.com/research/riema...



From anthropic.com

2:28 PM · Aug 10, 2026 · **10.2M** Views

x.com/AnthropicAI/status/2086867246073401655

Hugging Face Search models, datasets, users...

← Back to Articles

Anatomy of a Frontier Lab Agent Intrusion: A Technical Timeline of the July 2026 Incident

Published July 27, 2026

Update on GitHub

▲ Upvote 482 +476

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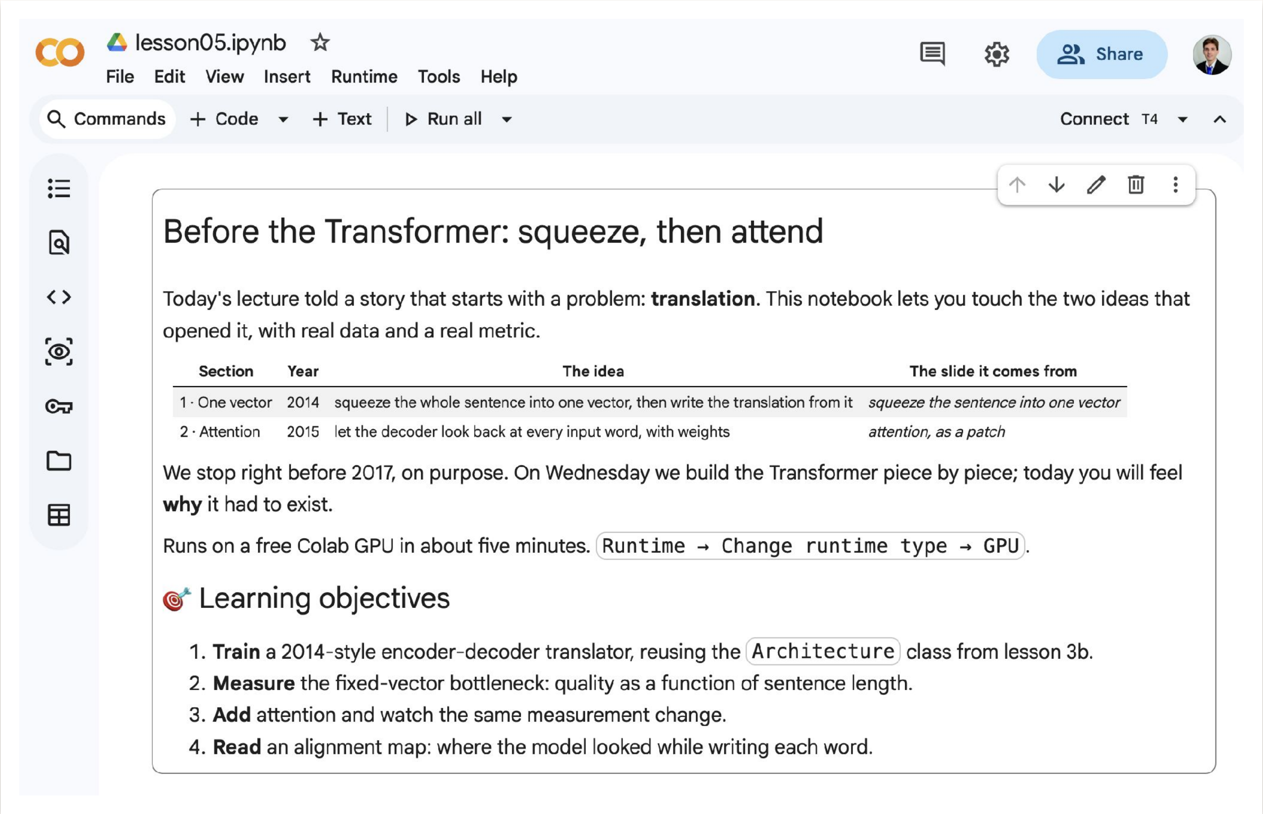
raphael g raphael-gl Follow

Christophe Rannou chris-rannou Follow

A companion technical writeup to our incident disclosure. This post walks through how the intrusion actually worked: the two initial-access vectors, how the agent pivoted and moved laterally, representative examples of the commands that were run and how we investigated with GLM 5.2 (an open-source model). Live credentials, internal hostnames, and specific indicators have been redacted or genericized, while the techniques are described exactly as observed by Hugging Face.

huggingface.co/blog/agent-intrusion-technical-timeline

Clone me!!!!



Before the Transformer: squeeze, then attend

Today's lecture told a story that starts with a problem: **translation**. This notebook lets you touch the two ideas that opened it, with real data and a real metric.

Section	Year	The idea	The slide it comes from
1 · One vector	2014	squeeze the whole sentence into one vector, then write the translation from it	*squeeze the sentence into one vector*
2 · Attention	2015	let the decoder look back at every input word, with weights	*attention, as a patch*

We stop right before 2017, on purpose. On Wednesday we build the Transformer piece by piece; today you will feel **why** it had to exist.

Runs on a free Colab GPU in about five minutes. `Runtime → Change runtime type → GPU`.

🎯 Learning objectives

1. **Train** a 2014-style encoder-decoder translator, reusing the `Architecture` class from lesson 3b.
2. **Measure** the fixed-vector bottleneck: quality as a function of sentence length.
3. **Add** attention and watch the same measurement change.
4. **Read** an alignment map: where the model looked while writing each word.

https://github.com/ivanovitchm/aiengineering